

IAN SNIDER

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Systems engineer seeking roles in autonomous systems and control.

EDUCATION

Graduate Coursework in Nuclear Engineering - University of California, Berkeley		Aug 2025 - Dec 2025
B.S. Mechanical Engineering - Washington University in St. Louis	4.00/4.00	May 2025
B.A. Physics, Math minor - Truman State University	3.89/4.00	Dec 2024

PROJECTS

Sniff **Consensus Protocol for Noisy Multi-Agent Formations** - *Developer* Fall 2025

- Consensus protocol for a random graph of agents perturbed with communication noise
- Used Erdos-Renyi graphs to generate adjacency matrices and applied communication noise with Random Matrix Theory
- Used ROS2, Python, and Rust, for design, simulations, and stability analysis

WashU Robotics **MateROV & MARINER Projects** - *Mechanical Lead* 2023 - 2025

- Designed and tested an autonomous underwater vehicle (AUV)
- Prototyped dive control using an IMU and depth state estimation with a Kalman Filter for 1D hovering
- Used Arduino for programming AUV sensors and control system
- Built a chassis and buoyancy engine

CrazyFlie 2.1 **cascaded PID/PD Control System** - *ESE 4481 Student* Spring 2025

- Designed a control system for a CrazyFlie 2.1 quadcopter and achieved stable hovering
- Used Simulink and MATLAB for simulating the control system response and finding stability margins
- Used C to program the quadcopter embedded system

EXPERIENCE

Lawrence Berkeley National Laboratory - *Lab Affiliate*, Berkeley, CA Aug 2025 - Dec 2025

- Developed a Monte Carlo simulation to optimize sensor timing cycles for event coincidence detection
- Studied detector unfolding and primary gamma extraction from the Oslo Matrix

Lawrence Livermore National Laboratory - *Graduate Intern*, Livermore, CA June 2025 - present

- Developed CRISP (Compute & Resource Intensive Simulation Pipeline), a Python tool for parallelizing and distributing high-performance computing (HPC) jobs
- Test driven Python development included Poetry, CI/CD, SQLite, MPI, CLI, feature branching, and Sphinx
- Used 2 world-class LLNL HPC clusters **Dane** and **Ruby** with Slurm sbatch/srun scripts
- CRISP successfully automated compilation and benchmarking workflows for ~3420 reactor simulation cases
- Maintainer of the CRISP project

Brookhaven National Laboratory - *Student Researcher*, Upton, NY Summers 2022 - 2024

- Developed BRR (Bayesian Resonance Reclassifier) for resonance reclassification on HPC clusters
- Applied machine learning using Scikit-learn to classify resonance spin-group assignments for capture cross-sections
- Used Random Matrix Theory and nuclear data for finding and scaling eigenvalues from reaction Hamiltonians to build synthetic training sets
- Used the transport code OpenMC and HPC clusters to parallel process perturbative sensitivity analyses for reactor models

Truman State University - *Student Astronomy Researcher*, Kirksville, MO 2021 - 2022

- Calculated trajectories for ~3000 Starlink satellites to optimize telescope viewing plans and developed a GUI

SKILLS

- **Programming:** C, C++, Python, Go, Bash, Lua, MATLAB, Mathematica, Octave, LaTeX, Typst, Vimscript
- **Software/Technical:** Simulink, SolidWorks, OpenMC, COG, MCNP, Pytorch, Scikit-learn, Slurm, Git, Linux, Arduino

- **Physics/engineering:** Autonomous Aerial Vehicle Control, Classical Mechanics, Electrodynamics, Electronics, Vibrations, Quantum Mechanics, Mathematical Physics, Nuclear Physics, Fluid Mechanics, Solid Mechanics, Acoustics, Materials Science, Thermal Systems, Criticality Safety, Radiation Biology, Turbojets, Ramjets
- **Math:** Linear Algebra, ODEs, Computing Structures, Control Systems, Machine Learning, Random Walks, Random Matrix Theory, Orthogonal Groups, and Optimizations

HONORS

Kenneth L. Jerina Prize for Outstanding Dual-Degree in Mechanical Engineering - WashU Engineering *April 2025*

- Honored for academic achievements and engineering community contributions

Conference Experience for Undergraduates 2023 - APS DNP *September 2023*

- Competitive research abstract award
- Invitation to present a research poster at the APS DNP Fall 2023 conference on The Big Island, HI

Conference Experience for Undergraduates 2022 - APS DNP *August 2022*

- Competitive research abstract award
- Invitation to the poster presentation at the APS DNP Fall 2022 meeting in New Orleans, LA

Sigma Pi Sigma Honor Society - Truman State University *May 2022*

- Recognized for service and academic scholarship in physics